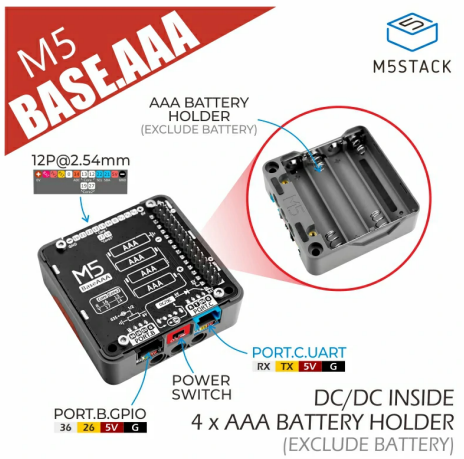
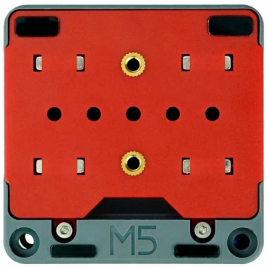
Base AAA

SKU:A122

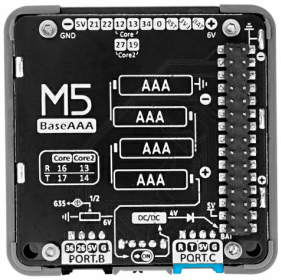




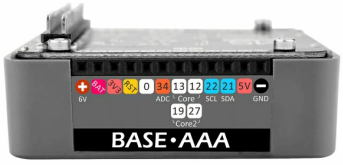












Description

Base AAA is a "5X5cm" series base powered by four AAA dry batteries. Unlike other lithium battery bases, **Base AAA** uses safe and reliable dry battery power supply. It also features an onboard physical power switch that can completely turn off the power to reduce dry battery loss. It has reserved two PORT.B and PORT.C interfaces for expansion, provides a 1 x 12P 2.54 pitch female header expansion interface, circular mounting holes on the side for easy fixing and expansion, supports magnetic installation at the bottom, and two embedded brass nuts for mounting.

Features

- 1 x 12P 2.54 interface
- PORT.B, PORT.C interface
- Power switch
- Supports two power supply modes:
  - Powered by four AAA dry batteries, stepped down from 6V -> 4V via DC-DC, supplied to the BUS BAT Pin
  - External power supply through Dupont wire to the BAT/GND of the 12P 2.54 interface
- Multiple fixation methods:
  - Bottom magnetic fixation
  - Two embedded brass nuts at the bottom
  - Three reserved mounting holes on the side

Includes

- 1 x Base AAA
- 1 x Hex Key L-Shape 1.5mm (For M2 Screw)

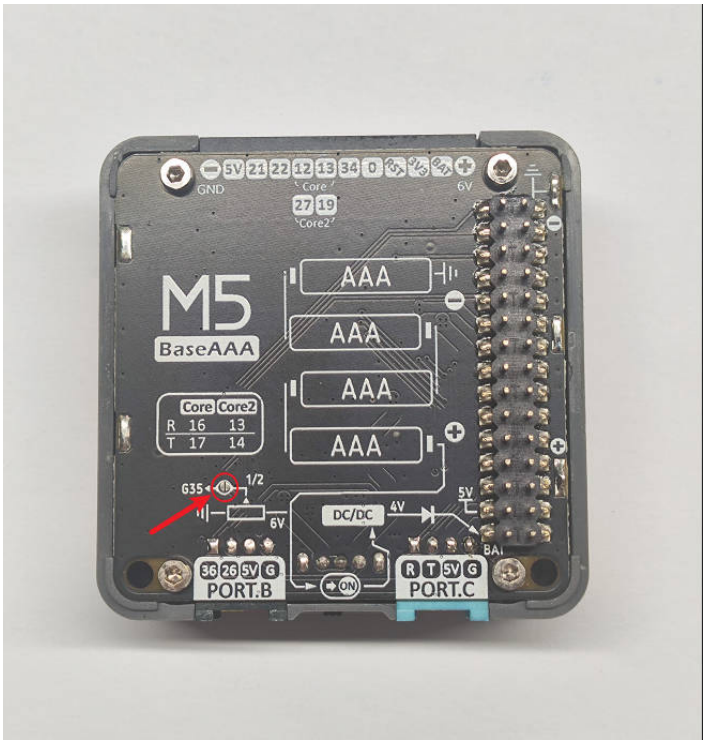
Specifications

Specification	Parameter
Product Size	54.0 x 54.0 x 21.0mm
Product Weight	30.0g
Package Size	80.0 x 55.0 x 28.0mm
Gross Weight	51.2g

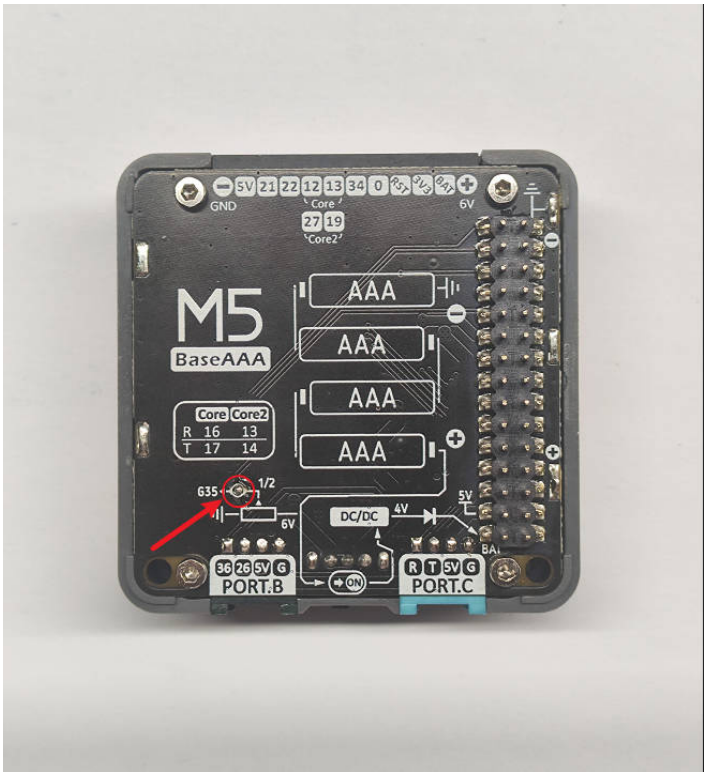
Learn

Soldering Tips

By default, the G35 pin is disconnected. If you need to read the power supply voltage, solder this pad as shown in the figure below:



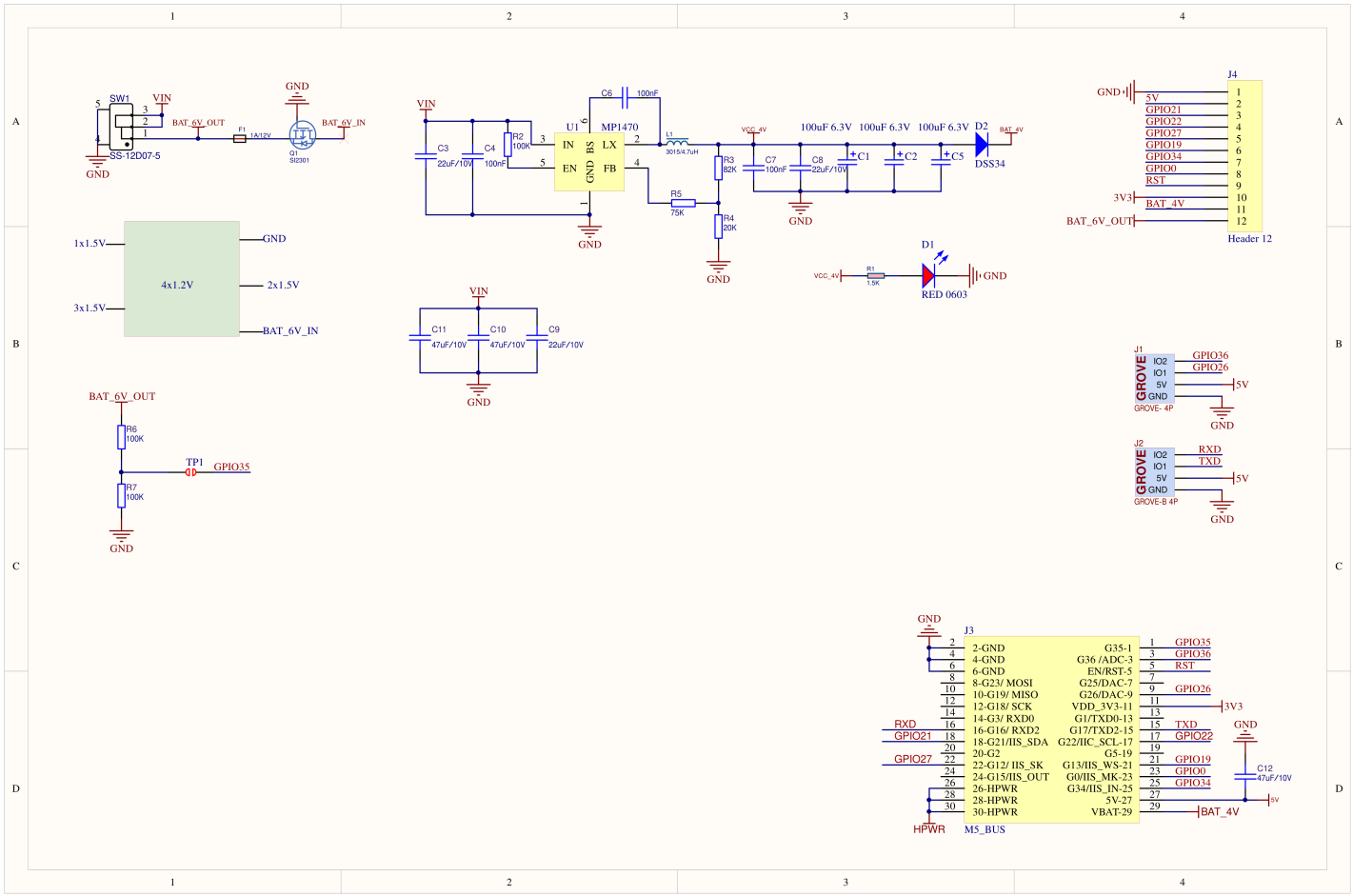
- Before soldering



- After soldering

Schematics

- Base AAA Schematics PDF



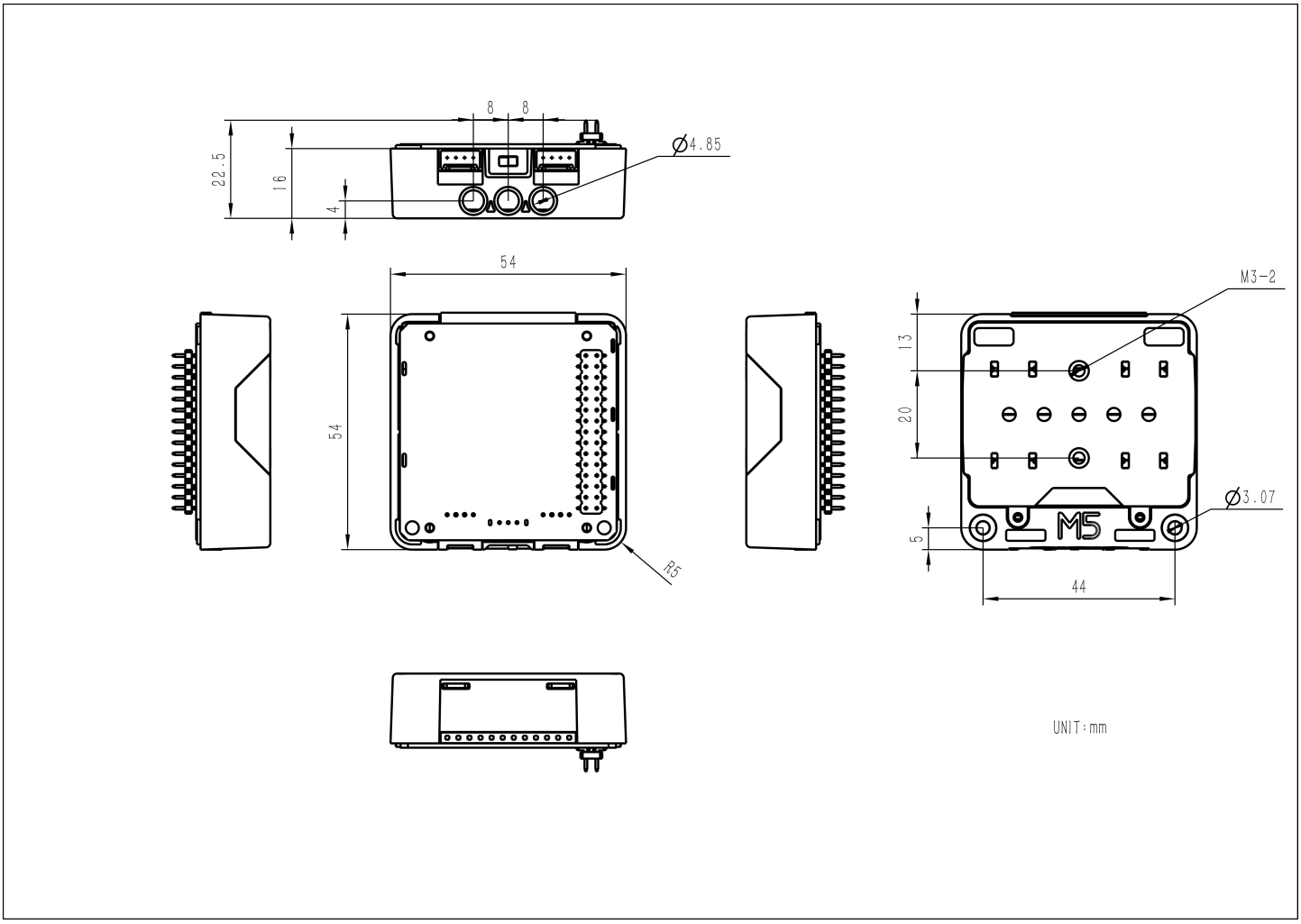
PinMap

M5-Bus

PIN	LEFT	RIGHT	PIN
GND	1	2	NC
GND	3	4	PORT.B
GND	5	6	RST
NC	7	8	NC
NC	9	10	PORT.B
NC	11	12	3V3
NC	13	14	NC
UART_RX	15	16	UART_TX
I2C_SDA	17	18	I2C_SCL
NC	19	20	NC
PIN	21	22	PIN
NC	23	24	PIN
HPWR	25	26	PIN
HPWR	27	28	5V
HPWR	29	30	BAT

Model Size

- Base AAA Model Size PDF



Softwares

Arduino

- Use ADC to read the voltage of the power pin

```
#include <M5Unified.h>
#define BAT_ADC_PIN 35
int16_t sensorValue = 0;
float voltage = 0.0;
void setup() {
  M5.begin();
  Serial.begin(115200);
  analogReadResolution(10);
}
void loop() {
  sensorValue = analogRead(BAT_ADC_PIN);
  voltage = (sensorValue * 3.0) / 1023.0;
  Serial.print("sensor = ");
  Serial.print(sensorValue);
  Serial.print("\t vol = ");
  Serial.print(voltage);
  Serial.println("V");
  delay(200);
}
```